

MAE 4730
10/1/2018
Lecture 16

Today:

- Review
- Constraints (cont'd)
- 2D rigid objects " $M = I\alpha$ "

Review

For any system

I. LMB: $\sum \vec{F}_i^{\text{ext}} = \underbrace{\sum m_j \vec{a}_j}_{m_{\text{tot}} \vec{a}_G}$

II. AMB_{IC}: $\sum \vec{M}_{i/C}^{\text{ext}} = \sum \vec{r}_{j/C} \times m_j \vec{a}_j$

any pt C.

- a) still in $\mathcal{F}$
- b) moves at constant velocity in $\mathcal{F}$
- c) COM = G = C

$$= \vec{r}_{G/C} \times m_{\text{tot}} \underbrace{\vec{a}_G}_{\vec{a}_{G/\mathcal{F}}} + \sum \vec{r}_{j/\mathcal{F}} \times m_j \vec{a}_{j/G}$$
$$= \dot{\vec{H}}_{G/C} + \dot{\vec{H}}_{/G}$$

$$= \frac{d}{dt} [\vec{H}_{G/C} + \vec{H}_{/G}]$$

$$\vec{H} = \vec{r} \times m \vec{v}$$

Goal: find motion and unknown forces

Method: solve ODE's

Need: ODES

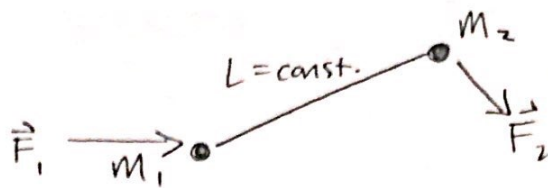
ODES: given state (position, velocity of system)

use I and II (or L.E) to find (state variables)

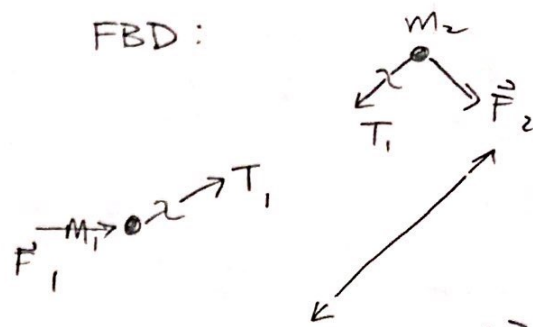
[& constant forces]

ex) 3D 2 masses connected by rod

DAE approach



FBD:



• why tension along rod?

- When connecting particles, can only have 2 force body

EOM.

$$\begin{aligned} \text{LMB: } \vec{F}_1 + T_1 \frac{\vec{r}_{12}}{|\vec{r}_{12}|} &= m_1 \vec{a}_1 \\ \vec{F}_2 - T_1 \frac{\vec{r}_{12}}{|\vec{r}_{12}|} &= m_2 \vec{a}_2 \end{aligned} \quad \left. \begin{array}{l} 6 \text{ eqns (3D)} \\ \text{in} \\ 7 \text{ unknowns} \end{array} \right\}$$

constraint eqn:

state variables: $\left. \begin{array}{l} \vec{r}_1, \vec{r}_2 \\ \vec{v}_1, \vec{v}_2 \end{array} \right\} 12$

$$|\vec{r}_{12}| = \text{constant}$$

→ save math by not dealing w/ square roots

$$|\vec{r}_{12}|^2 = \vec{r}_{12} \cdot \vec{r}_{12} = \text{constant}$$

$$\frac{d}{dt} \underbrace{(\vec{r}_{12} \cdot \vec{r}_{12})}_{(\vec{r}_2 - \vec{r}_1)} = 2 \dot{\vec{r}}_{12} \cdot \vec{r}_{12} = 0$$

$$\frac{d}{dt} \left\{ \dot{\vec{r}}_{12} \cdot \vec{r}_{12} \right\} = \boxed{\vec{r}_{12} \cdot \ddot{\vec{r}}_{12} + \dot{\vec{r}}_{12} \cdot \dot{\vec{r}}_{12} = 0} \quad \left. \begin{array}{l} 1 \text{ eqn} \\ \text{in} \\ 18 \text{ variables} \end{array} \right\}$$

unknowns: $\ddot{\vec{r}}_1$ & $\ddot{\vec{r}}_2$ (in prior eqn)

constraint eqn + 2 LMB eqn

$$\begin{bmatrix} \text{LMB eqn} \\ \text{const eqn} \end{bmatrix} \begin{bmatrix} \text{const eqn} \\ 0 \end{bmatrix} \begin{bmatrix} \dot{v}_1 \\ \dot{v}_2 \\ T_1 \end{bmatrix} = \begin{bmatrix} \text{all terms} \\ w/\ddot{\vec{r}}_1, \ddot{\vec{r}}_2 \\ \dot{v}_1, \dot{v}_2, \ddot{\vec{r}}_1, \ddot{\vec{r}}_2 \end{bmatrix} \star$$

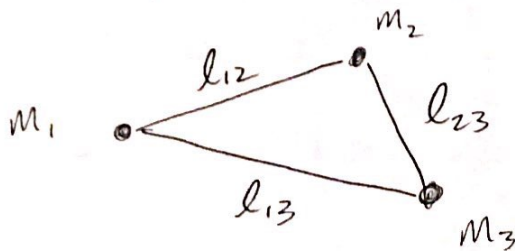
7 rows

7 col

Solve $\star$ inside r.h.s. file

THIS ex IS simplest rigid object

ex) 3 particles (2D)



$\vec{F} = m\vec{a}$ for each particle (6 eqns)

3 constraint eqns

(3 eqns)
9 eqns

ex) 27 particles in a rigid object

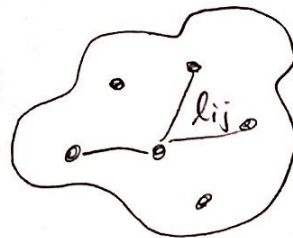
$\Rightarrow 27 \times 2$ LMB eqns

constraint eqns?

options:

- Connect all points: $27 \times 13 = 351$ const eqns X

• 51 $\checkmark$ (start w/ 2, add 1 particle at a time)



$\Rightarrow 51 + 54 = 105$ eqns in DAE matrix

Recall about constraints

- Have system
- then add constraints
- 2 choices

① Maximal coordinates

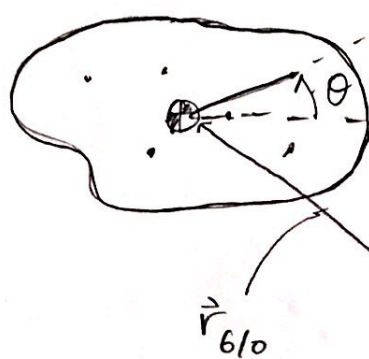
- Add constraint eqs to $\vec{F} = m\vec{a}$

get (maximal coordinates)

use for
rigid
object in
2D

② Minimal coordinates & a subset (projection)
of LMB & AMB eqs.
get (minimal coordinates)

Rigid Object in 2D:



- Lots of particles
(like 10^{23} of them)

θ : angle of some pair of points
we like & fix forever

Good minimal coordinates

know $\vec{r}_6, \theta$

Can calculate $\vec{r}_j$

$\dot{\vec{r}}_6, \dot{\theta}$

$\dot{\vec{v}}_j$

$\ddot{\vec{r}}_6, \ddot{\theta}$

$\ddot{\vec{a}}_j$

LMB:

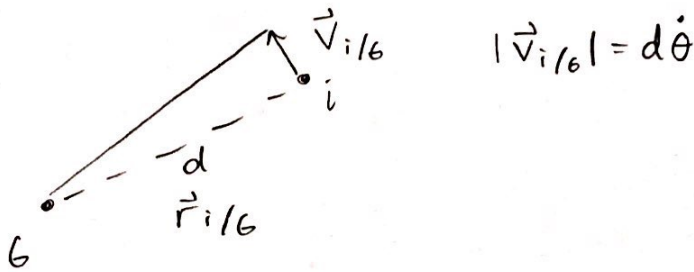
$$\sum \vec{F}_i^{\text{ext}} = m_{\text{tot}} \vec{a}_G$$

$\vec{r}$ are minimal coordinates

AMB:

$$\vec{H}_{/c} = \underbrace{\vec{r}_{G/c} \times m_{\text{tot}} \vec{v}_G}_{\text{}} + \underbrace{\sum \vec{r}_{i/c} \times m_i \vec{v}_{i/G}}_{(\star\star)}$$

$$(\star\star) \Rightarrow \sum \vec{r}_{i/G} \times m_i \vec{v}_{i/G} = \sum d_i \cdot m_i d_i \dot{\theta}$$



$$\vec{H}_{/G} = \dot{\theta} \underbrace{\sum m_i d_i^2}_{I^G}$$

I^G = moment of inertia w.r.t G

$$\vec{H}_{/c} = \vec{r}_{G/c} \times m_{\text{tot}} \vec{v}_G + \dot{\theta} I^G \hat{k}$$

$$\dot{\vec{H}}_{/c} \stackrel{\text{good point}}{=} \frac{d}{dt} (\vec{r}_{G/c} \times m_{\text{tot}} \vec{v}_G + \dot{\theta} I^G \hat{k})$$

$$\boxed{\sum \vec{M}_{/c}^{\text{ext}} = \vec{r}_{G/c} \times m_{\text{tot}} \vec{a}_G + \underbrace{I^G}_{\alpha} \ddot{\theta} \hat{k}}$$